

Empty Seats



Curse of Dimensionality

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Chameleon - Performance



Curse of Dimensionality



The higher the dimensionality and degrees of freedom of a model, the more data is needed to properly constrain it. The upside of flexibility and depth comes with a downside of data hunger and interpretative problems.



dimensionality, sparsity, concentration of measure/information, curse/blessing, kernel trick, regularization



- ▲ High-dimensional spaces allow flexible, expressive representations.
- ▼ Data covers fraction of the space; inference, generalization, interpretation fragile.
- ▶ Around target card, create empty space. Imagine what might occupy it.



"For which part of my model can I not estimate the amount of data needed to satisfy it?"



The theaters' main hall was so colossal, most members of the audience could barely hear the actors speak. The stories that reached their ears were fragmentary and confused.